Experiment 9.2

Simulation of Space Wave Propagation Characteristics using MATLAB

Name: Rohith Krishna R	**Reg. No:** 192312013

Aim

To simulate space wave (line-of-sight) propagation and analyse free space path loss, received power, LOS range and link budget.

Software Used

MATLAB

Theory / Formulas

Free space path loss: FSPL (dB) = 20 $\log_{10}$(d) + 20 $\log_{10}$(f) + 32.44, with d in km and f in MHz.
Received power: $P_r = P_t + G_t + G_r$ - FSPL (dBm).
The LOS range depends on the transmitting and receiving antenna heights.

Procedure

1. Open MATLAB and create a new script.
2. Define the parameters and generate the signals/data.
3. Compute FSPL and received power versus distance.
4. Compute the LOS communication range.
5. Plot the link budget analysis.

Output

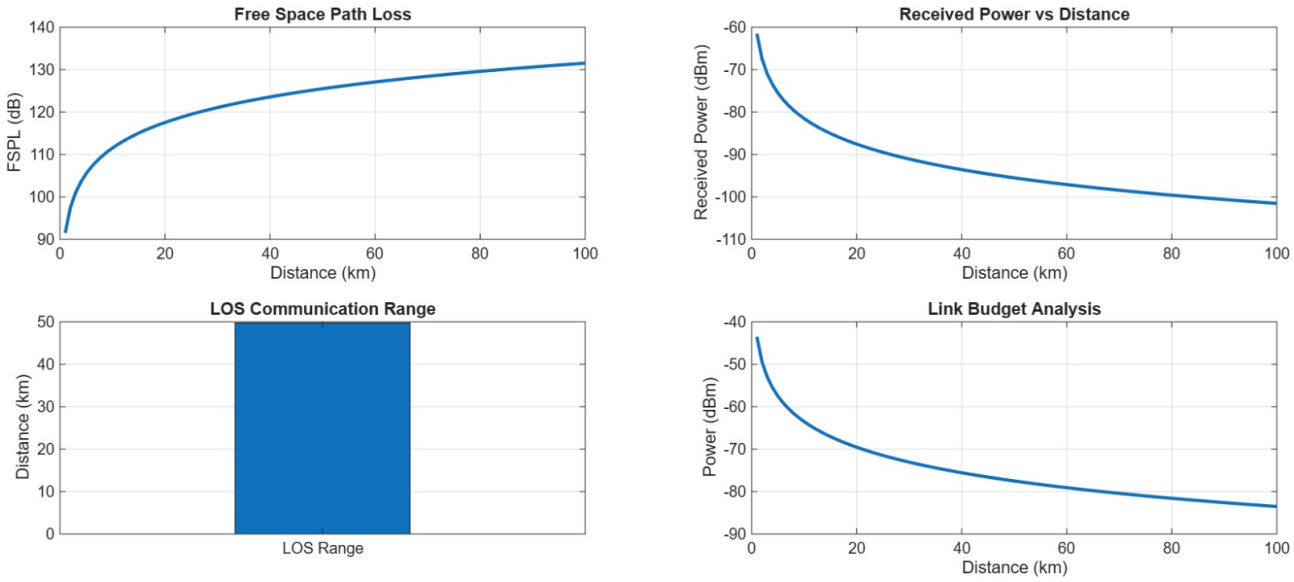


Fig. 1: Free space path loss, received power vs distance, LOS communication range and link budget analysis

Result

The free space path loss increases and the received power decreases with distance. The LOS range and the link budget were analysed for the space wave link.